25901231

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEECT202 - DATA SCIENCE FOR AI
(2024 ADMISSION ONWARDS)
SCHEME

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-MCQ
Answer all questions-Each question carries 1 marks

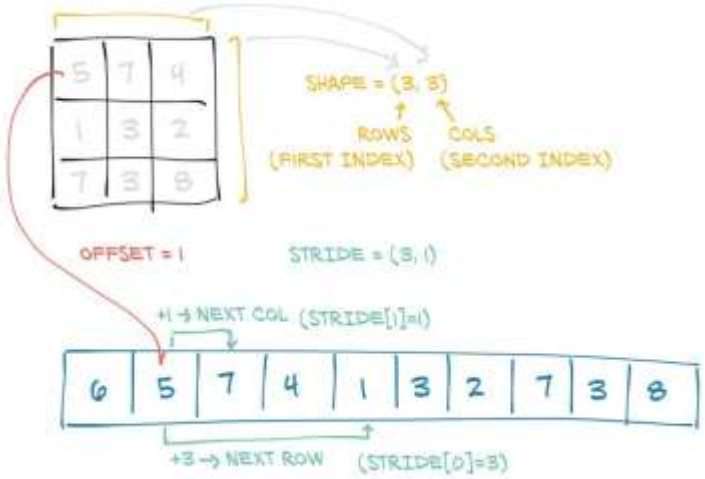
1. **d)** Faster parallel computation 1 mark
2. **c)** 1D is used for sequential data, 2D is used for images 1 mark
3. **b)** Vanishing gradient problem 1 mark
4. **a)** Leaks future info 1 mark
5. **c)** 32×32 1 mark
6. **b)** Numbers with fractional parts 1 mark
7. **c)** 5 1 mark
8. **b)** They retain contextual information from earlier words 1 mark
9. **c)** To align source and target parts in translation 1 mark
10. **c)** Encoder-decoder sequence tasks 1 mark
11. **b)** By improving readability 1 mark
12. **b)** Same and valid 1 mark
13. **c)** Generate new data similar to training data 1 mark
14. **b)** Yes, within one sequence 1 mark
15. **a)** Memory cell 1 mark
16. **d)** Physical memory where tensor data is stored 1 mark

Part B
Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

17. Explain the role of stride in memory-efficient tensor operations. 5 mark

Stride: The step size to move to the next element along each dimension.Stride tells how many memory steps you must move to go from one element to the next along each dimension. Important because tensors do not always store data in a contiguous way.

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18. Comprehend the concept of local connectivity in convolutional neural networks. 5 mark

Local Connectivity means:Each neuron in the convolutional layer is connected only to a small spatial region (called a receptive field) of the input.

For example:

- If the input image is 32 × 32
- A filter (kernel) size is 3 × 3
- Each output neuron looks at only 3 × 3 pixels, not the entire image.

Local Connectivity is Needed because Images have spatial structure:

- Nearby pixels are strongly related.
- Important features (edges, corners, textures) are local patterns.

Local connectivity helps the network to Capture local patterns , Reduce number of parameters, Improve computational efficiency, Prevent overfitting

19. Explain the concept of Recurrent Neural Networks (RNN) and distinguish them from Feed Forward Neural Networks.

Concept of RNN-2 mark

Concept of FFNN-2 marks

Difference(any 3)-1 mark

A Recurrent Neural Network (RNN) is a type of neural network designed to process sequential data by maintaining a memory of previous inputs.Unlike standard networks, RNNs have loops in their architecture.The output at time step *t* depends on:

- Current input xt
- Previous hidden state ht−1

RNNs “remember” past information, making them suitable for Language modelling,Speech recognition,Time-series prediction

A Feed Forward Neural Network is the simplest neural network where,Data flows in one direction only (input → hidden → output),There is no memory of previous inputs,Each input is processed independently

Difference Between RNN and FFNN

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20. Describe why scaling by sqrt(d_k) is used in attention overview. 5 mark

In attention mechanisms, the similarity between query (Q) and key (K) vectors is computed using a dot product.Attention Score=Q·K. This score is then passed through a softmax function to obtain attention weights.When the dimension of the key vector dk becomes large,The dot product Q·K can become very large,Large values fed into softmax cause,Extremely peaked distributions,Very small gradients during backpropagation,Training becomes unstable or slow.To fix this, the dot product is scaled:

21. Explain the role of channels in image tensors. 5 mark

Channels in image tensors represent different layers of information per pixel—from basic color (RGB) to advanced spectral bands (hyperspectral). They are essential for enabling neural networks to understand and process images effectively.In deep learning, an image is represented as a tensor with dimensions: Height×Width×Channels.Channels represent different types of information captured at each pixel.Grayscale Image → 1 channel (intensity only),RGB Image → 3 channels (R, G, B)

22. Explain how convolutional layers extract features from images. 5 mark

Convolutional layers extract features by applying learnable filters across the image, generating feature maps that highlight important patterns. By stacking multiple layers, CNNs build a hierarchy of features, enabling accurate image understanding.

- Take the input image (as a matrix of pixel values).
- Choose a small filter (kernel) (e.g., 3×3).
- Place the filter on the image (starting at top-left).
- Multiply corresponding values (image × filter).
- Add all values → gives one number.
- Store the result in the output (feature map).
- Slide the filter across the image and repeat.
- Form the full feature map after scanning entire image.
- Apply activation (ReLU) to keep important features.
- Use multiple filters to detect different features (edges, textures, etc.).

23. Explain many-to-one and many-to-many types of RNN 5 mark

Many-to-One → used when a whole sequence leads to one decision

Many-to-Many → used when each input or sequence produces multiple outputs

24. Explain step-by-step: from QKV projection to weighted sum in attention. 5 mark

Start with input embeddings-Convert each word/token into a vector.

Create Query, Key, and Value-From each input, generate three vectors:

Query → what the word is searching for

Key → what the word contains

Value → actual information of the word

Compare Query with all Keys-Each word checks how related it is to every other word.

Get attention scores-These comparisons produce scores showing importance.

Scale the scores-Adjust values to keep them stable during training.

Apply softmax-Convert scores into probabilities (weights that sum to 1).

Multiply weights with Values-Give more importance to relevant words and less to others.

Add (sum) the weighted values-Combine them to form the final output for each word.

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Part C

Essay Questions

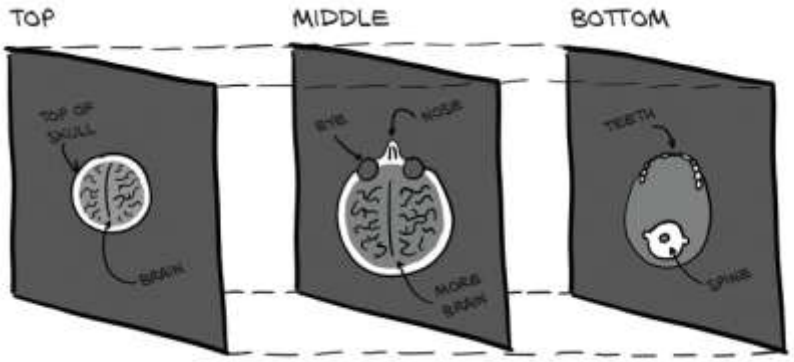
Answer any 2 questions

Each question carries 12 marks

25. Describe the role of volumetric data in modern machine learning applications.

While 2D images represent flat scenes captured by cameras, 3D images (volumetric data) arise in domains such as medical imaging, especially CT (Computed Tomography) scans. A CT scan consists of multiple 2D slices taken along the head-to-foot axis of the human body. When stacked together, these slices form a 3D volume that represents the internal anatomy.

In CT imaging, each pixel intensity corresponds to tissue density—from low-density regions like lungs (dark) to high-density structures like bones (bright). These values are computed from how X-rays are attenuated as they pass through the body and are reconstructed using mathematical algorithms. Unlike RGB images, CT data usually has only one intensity channel, similar to grayscale images.Because there is only a single channel, the raw CT data is often stored as a 3D array: D×H×W



where:

- **D** = depth (number of slices),
- **H** = height,
- **W** = width.

Here, the extra dimension represents physical depth in space, not color information.

Volumetric Data as Tensors.From a tensor perspective, volumetric data is not fundamentally different from image data—it simply has one extra dimension. When used in deep learning:

- A single volume is represented as **C × D × H × W**
- A batch of volumes is represented as:N×C×D×H×W

Medical images are commonly stored in **DICOM** format. Using tools like imageio.volread, a directory of DICOM slices can be loaded and automatically assembled into a 3D NumPy array. Since PyTorch expects a channel dimension, an additional dimension is added using unsqueeze to represent the single intensity channel.Once converted to a floating-point tensor, multiple CT volumes can be stacked along the batch dimension, enabling training and inference with **3D** convolutional neural networks.

26. Explain in detail the common layer patterns used in CNN architectures with suitable diagrams.

Explanation-6 marks

Diagrm-6 marks

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27. Describe the structure of an autoencoder and summarize its role in feature learning.

Autoencoder Explanation-7 marks

Diagram-3 marks

Role in feature learning explanation-2 mark

28. Explain how global and local attention are combined in long text processing.

Explain global attention-4 mark

Explain local attention-4 mark

Explain combined attention-4 mark

Local attention → like reading word by word
Global attention → like looking at important highlights
Together → full understanding of long text
Combining global and local attention allows models to process long text by balancing efficiency (local focus) and context awareness (global connections).